

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457094.80718 +/- 0.0009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.012 +/- 0.056
Transit depth (Rp/Rs)^2	0.014 +/- 0.13 [%]
Orbital Inclination (inc)	78.6 +/- 2.88
Airmass coefficient 1 (a1)	3.7 +/- 2.09
Airmass coefficient 2 (a2)	-0.88 +/- 0.79
Scatter in the residuals of the lightcurve fit is	59.06 %
Best Comparison Star	None
Optimal Aperture	4.04
Optimal Annulus	11.58
Transit Duration (day)	0.021 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.012 ± 0.056 vs 0.1594 (deviation 2.09σ)
Duration fit vs published	0.021 d vs 0.0512 d (deviation 1.33σ)
Mid-time O–C	1.3 min
Depth SNR	0.2
Residual scatter	59.06 %

Light curve

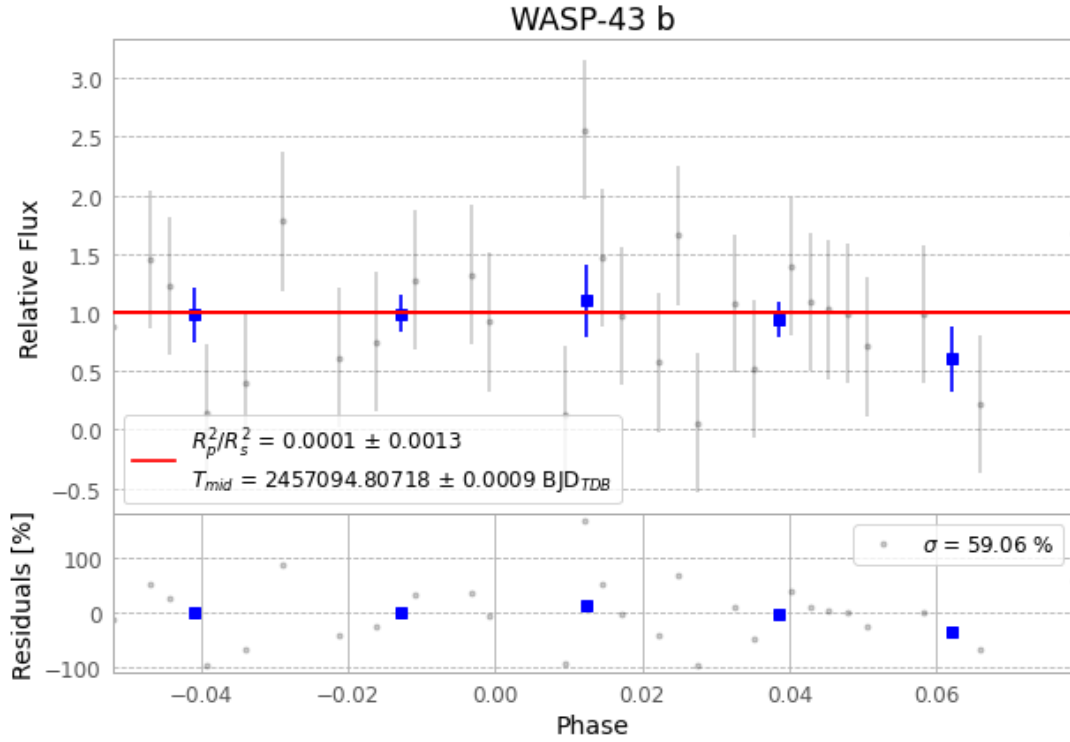


Figure 1 — FinalLightCurve_WASP-43 b_2015-03-12.png (SHA-256 976f85a2cff40983...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [171, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ff3efe6b9c7dd883...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

60 FITS frames (39 MB), WASP-43150313060012.FITS → WASP-43150313085712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150313.log (d190d25d4af0...), FinalLightCurve_WASP-43 b_2015-03-12.png (976f85a2cff4...), FinalLightCurve_WASP-43 b_2015-03-12.csv (7917ddb60a43...)

Compute resources

Wall time 154.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:21Z.